

Week 6: Options Pricing, the Volatility Surface, and Calibration

Building a volatility surface from messy crypto data

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Learning Objectives

- ▶ Recall the Greeks, including vanna and volga
- ▶ State the no-arbitrage constraints on the vol surface
- ▶ Calibrate SVI and SABR with attention to stability

Lecture 1: Greeks & the Surface

- ▶ Black-Scholes and the Greek vector
- ▶ Vanna, volga, and volatility risk
- ▶ The implied-vol smile and term structure
- ▶ Calendar and butterfly arbitrage

Lecture 2: SVI & SABR Calibration

- ▶ Raw SVI and the arbitrage-free formulation
- ▶ SVI calibration by nonlinear least squares
- ▶ The SABR formula and the Hagan expansion
- ▶ Time-stability of calibrated parameters

Recitation & Problem Set 6 (Python/PyTorch)

- ▶ Live calibration in PyTorch (autograd Jacobians)
- ▶ PS6: BS Greeks via autograd, SVI/SABR fits, stability study
- ▶ Reading: Gatheral (2006); Gatheral-Jacquier (2014); Hagan et al.